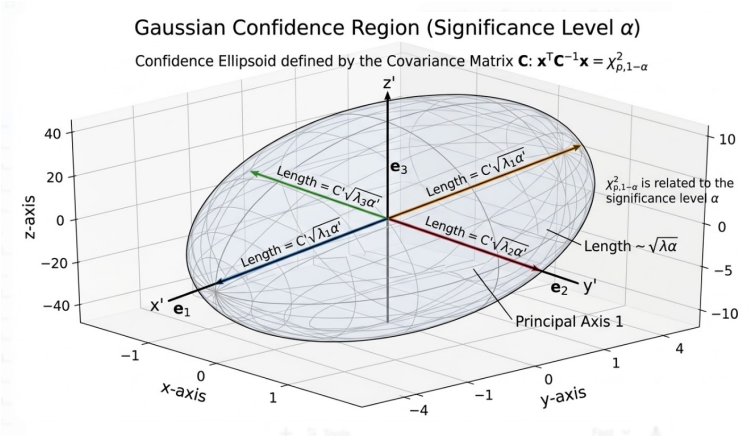
Principal axes of ellipsoid

Let C be the covariance matrix of size $n \times n$ of an n-dimensional Gaussian vector: $X = (X_1, X_2, \ldots, X_n)$. The covariance matrix can be proved to be positive definite by its definition. When samples are drawn from this distribution, their confidence region forms an n-dimensional ellipsoid. Below are three fundamental properties of this ellipsoid in $\mathbf{R}^n$:

- **Principal axes directions:** There are n principal axes and their directions are defined by the eigenvectors of C.
- **Principal axes lengths:** The semi-axis length along a specific direction is given by $\sqrt{\lambda\alpha}$, where λ is the eigenvalue of C corresponding to that axis's eigenvector. The parameter $\alpha > 0$ is set by the user, indicating that the ellipsoid can be scaled radially by increasing α.
- **Confidence level and probability:** To ensure that a sample X falls within the ellipsoid with a specific probability p, the scaling factor α must be set as:

  $$\alpha = F^{-1}_{\chi^2_n}(p),$$

  which is the inverse cumulative distribution function (CDF) of chi-square with n degree of freedom, evaluated at p.



Task

For a given covariance matrix, and a probability p, find the longest principal axis direction with unit norm and its semi-length that the Gaussian confidence region can cover the Gaussian samples with specified probability. Write a function named `ellipsoid_longest_axis(C,p)`.

Input Format

- $C \succ 0$ of size $n \times n$ where $2 \leq n \leq 1000$.
- $0 < p < 1$: a probability value.

Output Format

- $x \in \mathbf{R}^n$: The direction of the longest principal axis. Its magnitude is normalized to 1.
- `semilength` (positive scalar): The length of the longest principal axis.

Sample Input

```
np.random.seed(4)
n, p = 2, 0.9
C = np.random.randn(n,n)
C = C.T@C/2
```

Sample Output

```
The longest axis's direction is [-0.773  0.635]
The semi-length is 1.884
```

Implementation

Goal: Fill in the following function:

```
import numpy as np

n, p = 2, 0.9
C = np.random.randn(n,n)
C = C.T@C/2

def ellipsoid_longest_axis(C,p):

  return x, semilength
```

Make sure you import any library you use to solve the problem.

Hint

- Check `scipy.stats` for statistical functions.